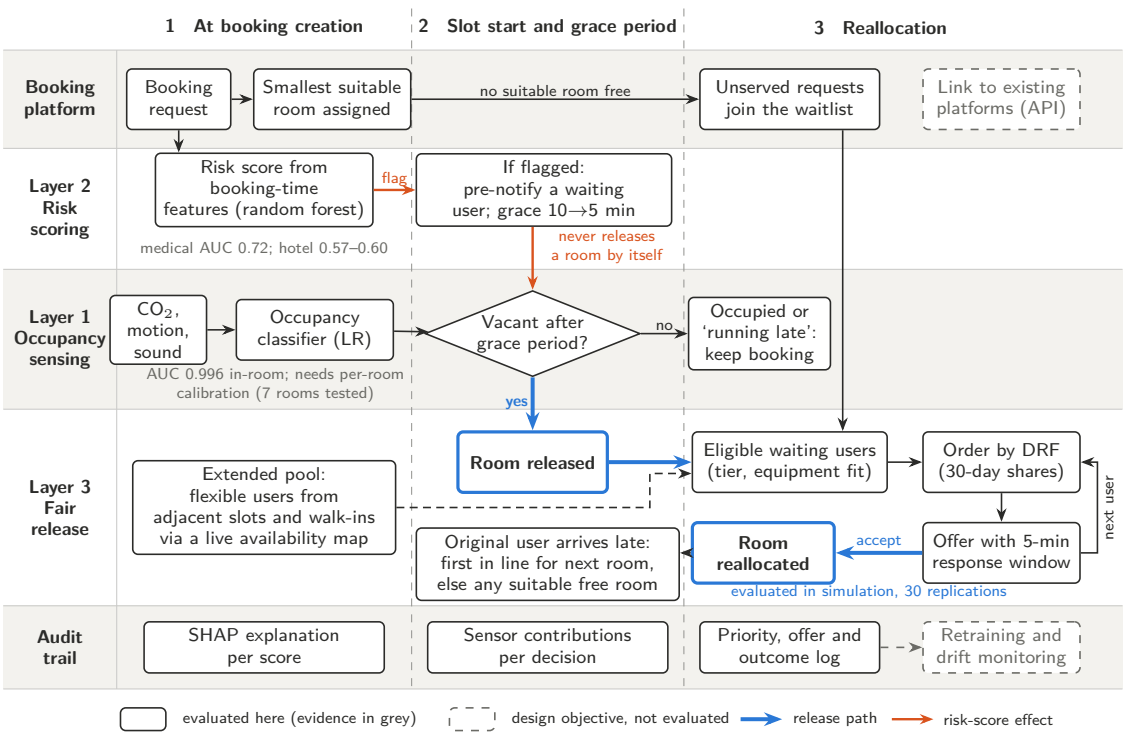

1 At booking creation
2 Slot start and grace period
3 Reallocation
Booking platform
Booking request
Smallest suitable room assigned
no suitable room free
Unserved requests join the waitlist
Link to existing platforms (API)
Layer 2 Risk scoring
Risk score from booking-time features (random forest)
flag
If flagged: pre-notify a waiting user; grace 10→5 min
never releases a room by itself
medical AUC 0.72; hotel 0.57–0.60
Layer 1 Occupancy sensing
CO2, motion, sound
Occupancy classifier (LR)
Vacant after grace period?
no
Occupied or 'running late': keep booking
yes
AUC 0.996 in-room; needs per-room calibration (7 rooms tested)
Layer 3 Fair release
Room released
Eligible waiting users (tier, equipment fit)
Order by DRF (30-day shares)
next user
Extended pool: flexible users from adjacent slots and walk-ins via a live availability map
Original user arrives late: first in line for next room, else any suitable free room
Room reallocated
accept
Offer with 5-min response window
evaluated in simulation, 30 replications
Audit trail
SHAP explanation per score
Sensor contributions per decision
Priority, offer and outcome log
Retraining and drift monitoring
evaluated here (evidence in grey)
design objective, not evaluated
release path
risk-score effect